Calculation of Basketball Scoring Probabilities

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Agenda

Results of basketball scoring calculations

- Overall Shooting Statistics
- Probabilities
- Conditional Probabilities - Future
- Conditional Probabilities - Retrospective

Overall Shooting Statistics

- **Overall probability of a make (result = TRUE): 0.4902**
- **Overall probability of a miss (result = FALSE): 0.5098**
- **Overall proportion of three-pointers (shot_type = 3): 0.5481**
- **Overall proportion of two-pointers (shot_type = 2): 0.4519**

Probabilities

- **What is the probability of Steph making 3 of the next 4 shots?**
  - $P(X = 3 \mid n = 4, p = 0.4902) \approx \mathbf{0.2402}$
- **What is the probability that 4 of the next 5 shots are three-pointers?**
  - $P(X = 4 \mid n = 5, p = 0.5481) \approx \mathbf{0.2039}$

What assumptions are you making?

The calculations for these probabilities rely on the Binomial Distribution and assume:

1. Shots are **independent events** (the result of one shot does not influence the next).
2. The **probability of success** (making a shot or taking a 3-pointer) remains **constant** for each trial (shot), based on the historical overall proportions.
3. The number of trials (n) is fixed (4 for the first, 5 for the second).
4. Each trial has only two outcomes (success/failure).

Conditional Probabilities - Future

- **If the next shot Steph shoots is a three-pointer...**
  - What is the probability he makes it?
    - $P(\text{make} \mid \text{3-pointer}) \approx \mathbf{0.4186}$
  - What is the probability it was taken while his team had the lead?
    - $P(\text{lead} \mid \text{3-pointer}) \approx \mathbf{0.4936}$
- **If the next shot Steph shoots is a two-pointer...**
  - What is the probability he makes it?
    - $P(\text{make} \mid \text{2-pointer}) \approx \mathbf{0.5772}$
  - What is the probability it was taken while his team had the lead?
    - $P(\text{lead} \mid \text{2-pointer}) \approx \mathbf{0.5154}$

Conditional Probabilities - Retrospective

1. What is the probability that it was a three-pointer? ($P(3 \mid \text{Made})$)

$$P(3 \mid \text{Made}) = \frac{P(\text{Made} \mid 3) \cdot P(3)}{P(\text{Made})}$$

$$P(3 \mid \text{Made}) = \frac{0.4186 \cdot 0.5481}{0.4902}$$

$$P(3 \mid \text{Made}) \approx \mathbf{0.4680}$$

2. What is the probability that it was a two-pointer? ($P(2 \mid \text{Made})$)

$$P(2 \mid \text{Made}) = \frac{P(\text{Made} \mid 2) \cdot P(2)}{P(\text{Made})}$$

$$P(2 \mid \text{Made}) = \frac{0.5772 \cdot 0.4519}{0.4902}$$

$$P(2 \mid \text{Made}) \approx \mathbf{0.5320}$$

(Verification: $0.4680 + 0.5320 = 1.0000$*)*